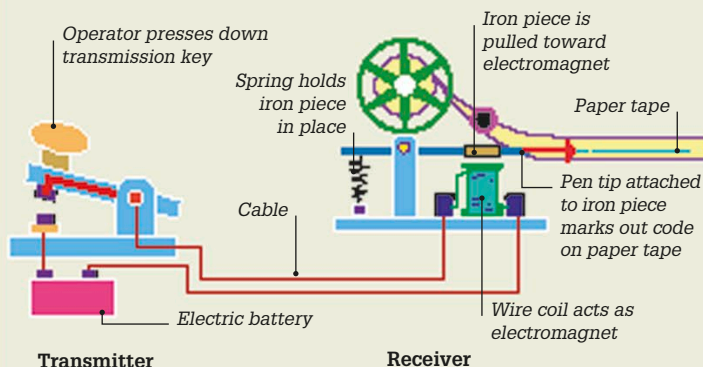


Morse telegraph

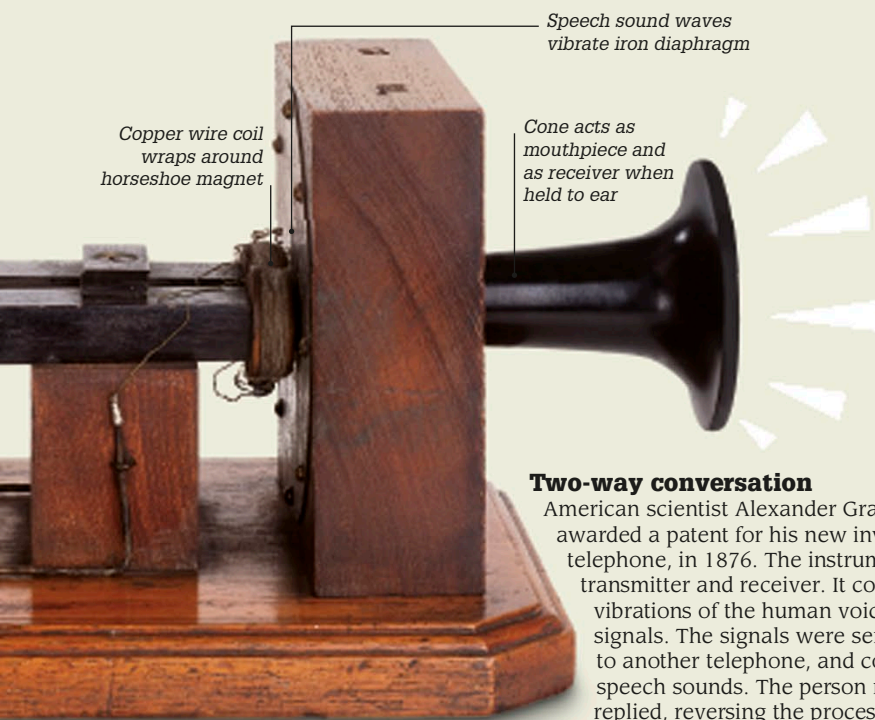
The electric telegraph sent messages, using Morse code (see p.127), as a series of electrical pulses representing the different letters of the alphabet. The operator pressed a key to transmit the pulses along a wire. At the receiving end, an electromagnet moved a pen that marked out the coded message on a paper tape.



Early Marconi radio receiver

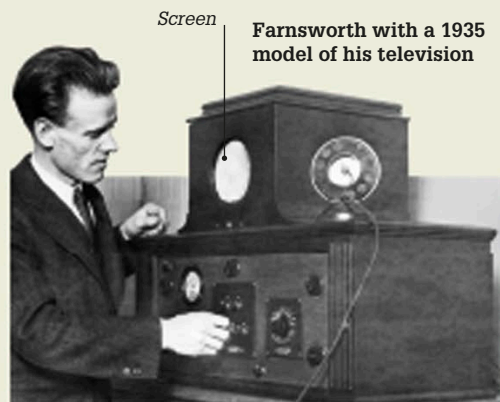
Radio at sea

Italian physicist Guglielmo Marconi invented the first long-distance radio (or wireless) communication system, which transmitted signals in Morse code. Used by ships at sea to send distress signals, it saved lives. The ocean liner *Titanic*, which sank in 1912, sent its last messages by Marconi radio, raising public awareness of the technology.



Two-way conversation

American scientist Alexander Graham Bell was awarded a patent for his new invention, the telephone, in 1876. The instrument was both transmitter and receiver. It converted the vibrations of the human voice into electrical signals. The signals were sent through a wire to another telephone, and converted back to speech sounds. The person receiving the call replied, reversing the process.



Electronic television

In 1927, two years after John Logie Baird's mechanical television, American inventor Philo T. Farnsworth built the first all-electronic television. It used a video camera tube (a type of vacuum tube) to capture images and transmit them as an electrical signal before reassembling them on a screen—a much faster process than Baird's use of a mechanical spinning disc.

1901–1902

A system designed by Irish-Italian physicist Guglielmo Marconi sent and received the first transatlantic radio signals, over a distance of more than 2,000 miles (3,300 km).

1925

Scottish inventor John Logie Baird transmitted the first moving image in a public demonstration of mechanical television.

1962

The first communications satellite, Telstar 1, was launched into space from Cape Canaveral, Florida, enabling television programs to be broadcast across the Atlantic.

1973

The first call was made on a handheld mobile phone. It weighed 2.4 lb (1.1 kg) and was the size of a brick.



First mobile phone